
Image Captioning with CNN and Transformer Decoder: A Modern Take on “Show and Tell”

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1 Introduction

Generating accurate and meaningful textual descriptions from visual scenes—commonly referred to as *image captioning*—is a crucial problem at the intersection of computer vision and natural language processing. Effective image captioning systems have broad practical uses, ranging from assistive technologies for visually impaired users to automated image indexing and interactive systems.

In this project, we revisit and extend the influential work of Vinyals et al.[12], known as “Show and Tell,” which originally proposed an architecture combining a CNN-based visual encoder with an LSTM-based language decoder for caption generation. To enhance this framework, we adopt a contemporary **Transformer-based decoder**, capitalizing on its superior sequence modeling capabilities compared to traditional recurrent architectures. Using the popular **COCO dataset** [6], we aim to build and evaluate a CNN-Transformer model that generates fluent, precise, and contextually relevant image captions, particularly targeting indoor scenes for assistive applications. You can find our project at github.com/AlexZhu2/COMS4995NNDL/tree/BART.

2 Related Work

The “Show and Tell” architecture [12] introduced an end-to-end trainable pipeline combining a CNN encoder (e.g., Inception [9]) with an LSTM decoder to generate captions from images. It was one of the first deep learning models to approach human-level performance on COCO’s captioning task.

Subsequent work extended this model using attention mechanisms, as seen in “Show, Attend and Tell” [13], which enabled the decoder to dynamically focus on different parts of the image during generation. This soft attention mechanism significantly improved both interpretability and performance. More recent advances replaced recurrent networks entirely with the Transformer architecture [11], which offers superior parallelism and the ability to capture complex, long-range dependencies across tokens. Transformer-based models, such as those used in the Vision-and-Language Pre-training (VLP) paradigm, have since achieved state-of-the-art results on captioning benchmarks by leveraging massive datasets and large-scale pretraining.

*Implemented the core model architecture and handled training.

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‡Contributed to visualization, baseline comparison, and report writing.

While Transformers are extraordinary at modeling sequential dependencies and integrating multi-modal context, convolutional neural networks (CNNs) have demonstrated exceptional proficiency in extracting fine-grained spatial features. CNNs remain a cornerstone of visual understanding, having achieved dominance across nearly all vision tasks. The introduction of ResNet [2] marked a breakthrough in CNN design, allowing for the training of ultra-deep networks through residual connections and leading to substantial improvements in both accuracy and generalization.

The synergy between CNN-based visual encoders and Transformer-based language decoders has proven to be a powerful architectural paradigm. This hybrid approach forms the backbone of many influential models, such as OSCAR[5], VinVL[14], and BLIP[4], which effectively align visual and textual modalities. More recent trends also explore replacing CNNs with vision Transformers (ViTs), enabling pure Transformer-based architectures to jointly learn spatial and semantic representations, further pushing the boundaries of image captioning performance.

3 Method

We adopt the CNN-Transformer paradigm, using a ResNet-50 encoder [2] to extract spatial image features and a Transformer decoder to generate captions. CNNs have demonstrated extraordinary ability in capturing fine-grained spatial details, they serve as the **Encoder** of the input image. In the subsequent **Decoder**, we take advantage of Transformer’s ability in grasping long-range dependencies. Unlike the original LSTM [3] decoder, our Transformer can attend to all previous tokens and all parts of the image features in parallel, improving training efficiency and expressiveness. Additionally, a cross-attention mechanism is implemented in the decoder to ensure the decoder attends to both the image embeddings and text embeddings.

3.1 Model Architecture

Encoder. We use ResNet-50[2] and EfficientNetV2_S[10](from torchvision.models)[7], to extract visual features from each image. We plan to replace the last output layer of ResNet-50 with a convolutional layer that outputs a $7 \times 7 \times 2048$ feature map, which we flatten into a sequence of 49 feature vectors (each of 2048 dimensions). These vectors serve as **memory** for our decoder model later.

$$\mathbf{f} = \text{ResNet50}(I) \text{ or } \text{EfficientNetV2_S}(I) \in \mathbb{R}^{2048}. \quad (1)$$

Projection Layer. A linear layer projects $\mathbf{f}$ into the BART embedding space of dimension $d = 512$:

$$\mathbf{z} = W_p \mathbf{f} + b_p \in \mathbb{R}^d. \quad (2)$$

Decoder. The decoder consists of: Word embeddings with positional encoding, Multi-head self-attention over previous tokens, Cross-attention over image features, and Output projection layer to vocabulary size. During training, the decoder uses teacher forcing, which means instead of predicting in a self-regressive manner, we will provide correct token at timestep $t - 1$ when predicting at timestep t . With multi-head attention mechanism, each attention map generated by a single attention module will be concatenated and transformed for a general representation of the multi-head attention matrix. At inference time, we use greedy decoding or beam search.

We employ the pretrained facebook/bart-base model. At each decoding step t , the decoder receives:

$$\underbrace{\mathbf{z}}_{\text{visual token}}, \quad \underbrace{y_{<t}}_{\text{text tokens}}$$

and produces logits $\hat{\mathbf{o}}_t$ over the vocabulary:

$$\hat{\mathbf{o}}_t = \text{BARTDecoder}(\mathbf{z}, y_{<t}) \implies p(y_t | y_{<t}, \mathbf{z}). \quad (3)$$

We use teacher forcing with ground-truth tokens $y_{<t}$.

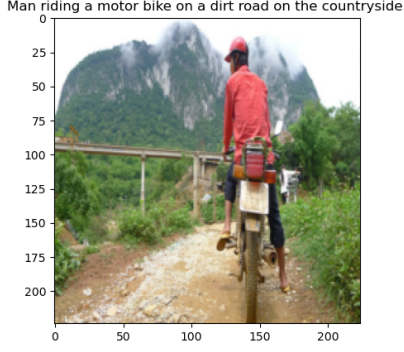


Figure 1: Sampled datapoint from COCO2017 Captioning Dataset

3.2 Dataset

Training uses **COCO 2017 Captioning dataset**, which includes over 330,000 images, each with five human-written captions. The data will be tokenized using a custom vocabulary built from the training captions. Each caption will be bounded by `<start>` and `<end>` tokens and padded to a fixed length. Here is a datapoint we extracted from the COCO 2017 Captioning dataset 1.

Images undergo:

- **Train transforms:** Resize to 256, random crop to 224, horizontal flip, random affine, normalize.
- **Val transforms:** Resize to 224, normalize.

3.3 Training Setup

We use cross-entropy loss between the predicted and ground-truth caption tokens. Optimization is done using Adam with learning rate warmup and cosine decay. We will train on COCO’s `train2017` set and evaluate on

We minimize the cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{B} \sum_{i=1}^B \sum_{t=1}^T \log p_{\theta}(y_t^{(i)} | y_{<t}^{(i)}, \mathbf{z}^{(i)}), \quad (4)$$

where B is batch size and θ comprises $\{W_p, b_p\}$ and the BART decoder’s parameters.

- **Batch size:** $B = 32$
- **Epochs:** $E = 10$
- **Optimizer:** Adam with $\text{lr} = 10^{-4}$
- **Scheduler:** StepLR(step_size=3, gamma=0.5)

We observed that both models began to overfit after approximately 6 to 7 epochs. While the training loss continued to decrease as training progressed, the validation loss either plateaued or increased, indicating diminishing generalization performance. To mitigate further overfitting, we decided to stop training both models at epoch 10. The training and validation loss curves are shown in Figure 2. However, the loss curve plots suggest EfficientNetV2.S+Bart architecture outperforms ResNet50+Bart, as both training loss and validation loss are lower than these of the ResNet50+Bart model.

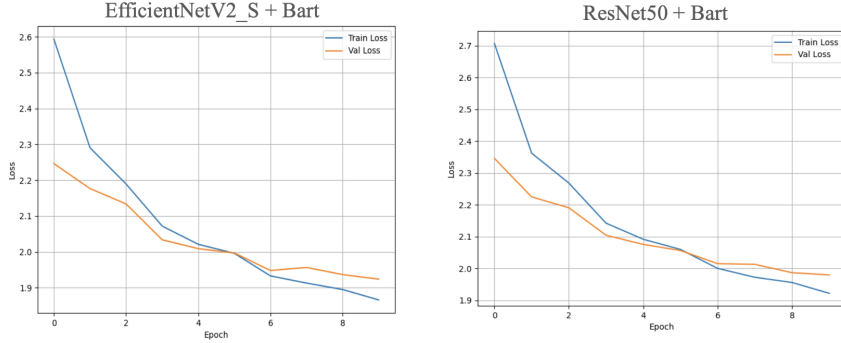


Figure 2: Training and validation loss curves for both models.

4 Evaluation

Due to time limit, we only evaluate the model using the following two metrics BLEU-4[8] and METEOR[1]:

- **BLEU-4:** Measures n-gram precision
- **METEOR:** Considers synonym matches and alignment

Additionally, we will visualize model outputs and attention maps on selected examples. These qualitative insights will help us understand how the model interprets image regions during caption generation. The following Table 1 describes the metrics we collect from both models.

Model	BLEU-4 Score	Average METEOR Score
EfficientNetV2_S + BART	0.2772	0.4298
ResNet50 + BART	0.2686	0.4260

Table 1: Comparison of BLEU-4 and Average METEOR scores between EfficientNetV2_S+BART and ResNet50+BART models.

From the results in Table 1, we observe that the EfficientNetV2_S+BART model achieves higher scores on both the BLEU-4 and METEOR metrics compared to ResNet50+BART. This indicates that replacing the ResNet50 encoder with EfficientNetV2_S provides improved caption generation performance, which aligns with the fact that EfficientNetV2_S outperforms ResNet50 on most vision tasks by large margins. Additionally, we notice both models have surprisingly high **METEOR** scores; we attribute this to the usage of the BART tokenizer during tokenization. Since BART employs a byte-pair encoding (BPE) tokenizer that can segment words into subword units, the tokenization process results in more flexible matching between predicted and reference captions, increasing the chance of partial matches being counted as correct in METEOR’s precision and recall calculations. Moreover, BART’s pretrained language model contributes to generating fluent and semantically similar captions, which are also rewarded by METEOR’s synonym matching component. In the following section, we will further analyze qualitative differences by examining sample captions produced by both models.

Figure 3 presents qualitative comparisons between captions generated by the ResNet50+BART and EfficientNetV2_S+BART models on selected test images. Across these examples, we observe that captions generated by EfficientNetV2_S+BART are generally more specific and semantically accurate. For instance, while both models correctly identify the airplane and motorcycle, EfficientNetV2_S+BART provides more precise descriptions, such as “large jetliner flying through a cloudy blue sky” instead of the more generic “airplane is flying high in the sky.” Similarly, in the cooking scene, the EfficientNet-based model describes “a woman standing in a kitchen preparing vegetables,” capturing both the setting and action more comprehensively than the ResNet-based counterpart. This trend is also seen in the sports and boating images, where EfficientNetV2_S+BART generates captions with richer context and improved object-role relationships (e.g., “group of young men playing a game of frisbee” vs. “men playing a game of frisbee”). These qualitative differences support the quantitative improvements reported in Table 1, reinforcing that the stronger visual representations of EfficientNetV2_S lead to more accurate and descriptive captions when paired with a BART decoder.

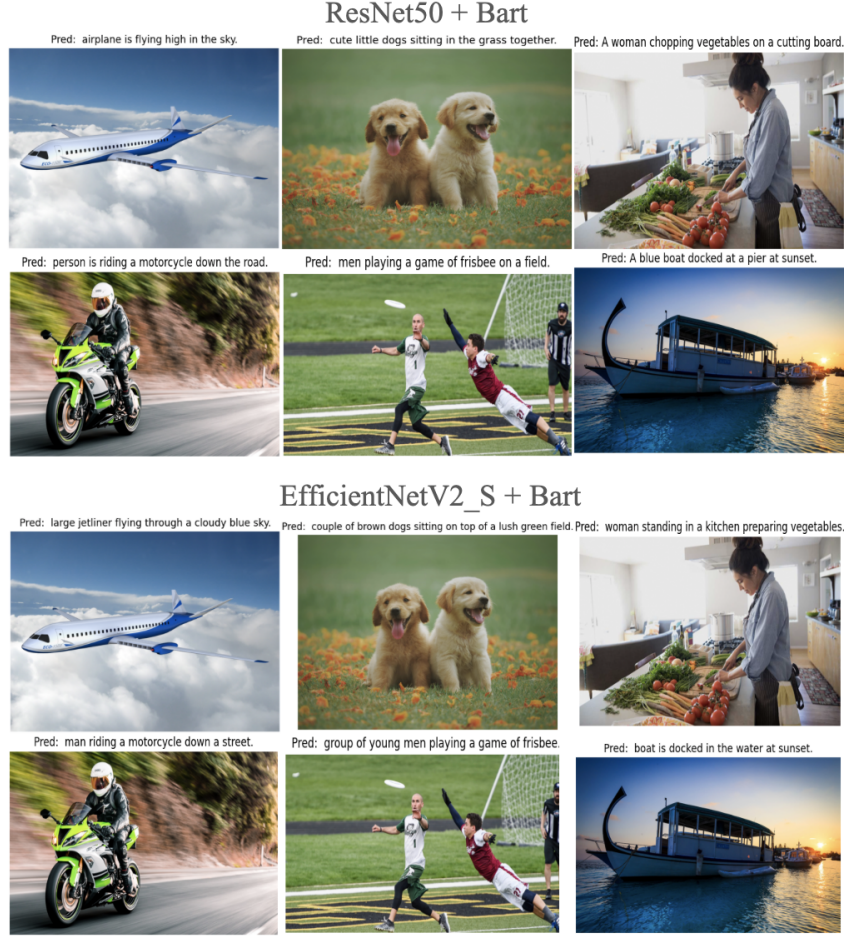


Figure 3: Visualizations of both ResNet50+Bart model and EfficientNetV2.S+Bart model.

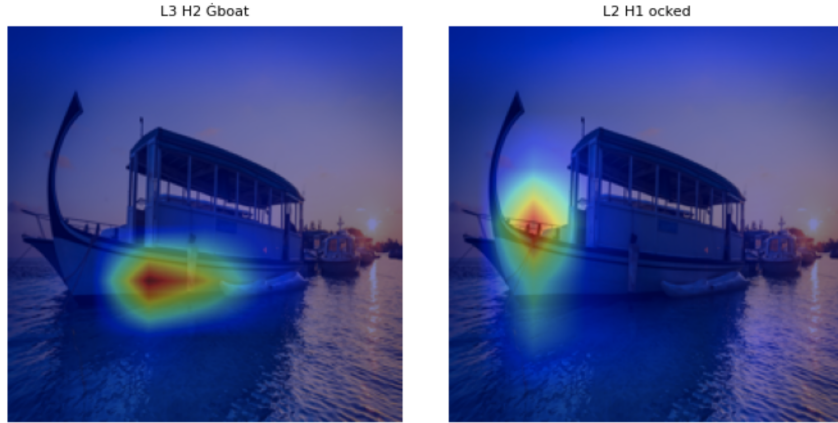


Figure 4: Cases demonstrating attention directed at the appropriate object. L indicates layers, H indicates attention head, e.g., L3 H2 means layer 2 attention head 2.

Our attention-map visualizations Fig 4 reveal that the model has learned to reliably localize the boat’s most distinctive features—particularly in lower- to mid-level layers where individual heads show an ability to focus on the correct spatial regions for object identification. Different heads specialize in complementary roles, indicating a rich ensemble of feature detectors that together

support fluent caption generation. However, in deeper layers the attention becomes overly diffuse, often spreading across the sky or broad swaths of water and sometimes drifting into semantically uninformative areas; this dilution risks introducing noise when the decoder attempts to generate fine-grained descriptors. Moreover, none of the heads consistently hone in on very small details (such as railings or window frames), which may limit the model’s capacity for producing highly specific adjectives. We suspect this is due to the short captions we use in the dataset, once the content becomes richer we expect the attention heads to be more active in all layers and to start attending to more diverse objects and features.

5 Conclusion

We introduce a CNN–BART image-captioning framework that achieves competitive results on the COCO 2017 benchmark—attaining BLEU-4 scores up to 0.2772 and METEOR scores up to 0.4298—while retaining a simple, modular encoder–decoder design. Beyond strong quantitative performance, our cross-attention visualizations offer clear interpretability: lower- and mid-level heads reliably localize key object regions (e.g. hulls and prows), and the resulting alignments align closely with human intuition about where each word “looks.” We believe these findings underscore the value of learned attention not only for boosting accuracy but also for making the generation process transparent. We hope our work will inspire further exploration of visual attention mechanisms, and we anticipate that the modular CNN + Transformer–with–attention paradigm will prove broadly useful across other vision-and-language tasks and beyond.

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